

AN ENGINEERING MAJOR PROJECT REPORT
ON
SMART HELMET FOR ACCIDENT PREVENTION AND
DETECTION

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COSMOS COLLEGE OF MANAGEMENT & TECHNOLOGY

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ABSTRACT

Road accidents involving two-wheelers are one of the major causes of fatalities and serious injuries in Nepal. Factors such as drunk driving, non-usage of helmets, over speeding, and delayed emergency response significantly increase the risk to motorcyclists. Traditional helmets provide only passive protection and fail to offer real-time accident detection or communication capabilities.

This project proposes an Smart Helmet system aimed at enhancing rider safety through real-time accident detection, prevention of riding under the influence of alcohol, and immediate transmission of emergency alerts to predefined contacts. The system integrates an ESP32 microcontroller, MPU6050 accelerometer and gyroscope, MQ-3 alcohol sensor, IR-based helmet wearing detection, NEO-10M GPS module, and A7670C GSM module. The helmet ensures that the motorcycle ignition is enabled only when the rider is wearing the helmet and is not under the influence of alcohol.

In the event of an accident, the system automatically detects abnormal motion patterns and sends the rider's real-time GPS location to predefined emergency contacts via SMS. The proposed system is cost-effective, reliable, and suitable for developing countries like Nepal, where quick medical response is often unavailable. Experimental results indicate improved accident response time and enhanced safety compliance, making the smart helmet an effective solution for reducing motorcycle-related injuries and fatalities.

Keywords: *Smart Helmet, Accident Detection System, Accident Prevention, Embedded System, Real-time monitoring, Rider Safety.*

PREFACE

In an increasingly technology-driven world, road safety has emerged as a critical public concern, particularly in developing countries where two-wheeler transportation is widespread. Motorcycle riders remain among the most vulnerable road users, facing heightened risks due to factors such as alcohol-impaired riding, inadequate helmet usage, poor road infrastructure, and delayed emergency response. These challenges demand intelligent, real-time safety solutions that go beyond conventional protective equipment.

This report presents the design, development, and evaluation of an Internet of Things (IoT)based Smart Helmet for Accident Prevention and Detection. By leveraging the sensing, processing, and communication capabilities of IoT technologies, the proposed system transforms a traditional helmet into an intelligent safety device. The system integrates motion sensors, alcohol detection, helmet-wear verification, and GPS-based location tracking, combined with GSM communication to enable real-time accident detection and automatic emergency alert transmission.

The primary aim of this work is not only to demonstrate the technical feasibility of such a smart helmet system but also to explore its practical applicability as a preventive and responsive safety tool. The proposed solution emphasizes affordability, modularity, and reliability, making it suitable for deployment in regions like Nepal, where rapid medical assistance is often limited and conventional safety enforcement mechanisms are insufficient.

This report provides a comprehensive account of the methodology adopted, the system architecture designed, the challenges encountered during implementation, and the results obtained through testing and evaluation of the prototype. The work highlights how embedded systems and IoT-based communication can be effectively utilized to reduce emergency response time, encourage responsible riding behavior, and ultimately minimize the severity of road accident outcomes.

It is hoped that the concepts, design approach, and findings presented in this report will contribute to future research and development in intelligent transportation safety systems.

The authors also anticipate that this report will serve as a valuable reference for students, researchers, engineers, and policymakers interested in the application of IoT and smart technologies toward improving road safety and saving human lives.

LIST OF ABBREVIATIONS

ADC	Analog to Digital Converter
DAC	Digital to Analog Converter
GPS	Global Positioning System
GSM	Global System of Mobile communication
IDE	Integrated Development Environment
IR	Infrared
I2C	Inter-Integrated Circuit
LED	Light Emitting Diode
PCB	Printed Circuit Board
SCL	Serial Clock Line
SDA	Serial Data Line
SMS	Short Message Service
RX	Receiver
TX	Transmitter

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1. INTRODUCTION

1.1 Overview

The immediate future presents an increasing concern regarding road safety in Nepalese urban and semi-urban regions due to rapid urbanization, growing vehicle density, poor traffic management, and risky riding behaviors such as drunk driving and non-usage of helmets. Motorcycle riders are particularly vulnerable to severe injuries and fatalities, often exacerbated by delayed emergency response and lack of real-time accident reporting. Long-term exposure to unsafe riding practices and inadequate safety mechanisms contributes significantly to preventable loss of life and serious physical trauma. Therefore, an effective and intelligent rider safety system is essential for accident prevention, timely medical intervention, and public safety awareness.

Traditional helmet systems provide only passive protection and lack real-time monitoring, accident detection, or communication capabilities. Existing advanced safety systems are often expensive, complex, and unsuitable for large-scale deployment in developing countries like Nepal. This project presents a low-cost, energy-efficient, and scalable Smart Helmet system aimed at improving rider safety through accident detection, drunk-and drive prevention, and instant emergency alert mechanisms.

The proposed system integrates an MPU6050 accelerometer and gyroscope for detecting sudden impacts and abnormal motion, an MQ-3 alcohol sensor for identifying alcohol consumption, and an IR-based helmet-wear detection sensor to ensure proper helmet usage. These sensors are interfaced with an ESP32 microcontroller, which processes the collected data and controls system operation. A NEO-M10 GPS module is used to obtain real-time location data, while a A7670C GSM module transmits emergency alerts containing location information to predefined contacts.

The use of IoT and cellular communication enables reliable real-time monitoring and rapid emergency notification with minimal power consumption, making the system suitable for both urban and remote regions of Nepal. Furthermore, the collected data can be utilized to analyze accident patterns, promote responsible riding behavior, and assist authorities in improving road safety policies. This system demonstrates a practical application of smart

technology toward reducing motorcycle-related injuries and fatalities and enhancing overall transportation safety.

1.2 Problem Statement

Existing motorcycle safety systems and conventional helmets provide only passive protection and lack intelligent features for accident prevention and post-accident response. Most traditional helmets do not include mechanisms for real-time accident detection, rider condition monitoring, or emergency communication. In many cases, accident victims fail to receive timely medical assistance due to the absence of automated alert systems and real-time location tracking, particularly in remote or less populated regions of Nepal.

Current advanced rider safety solutions often rely on complex architectures, high-cost hardware, or continuous internet connectivity, making them unsuitable for large-scale adoption in developing countries. Additionally, many existing systems do not effectively address critical preventive factors such as riding under the influence of alcohol and non usage of helmets. The lack of integrated drunk-and-drive prevention, helmet-wear verification, and reliable communication mechanisms significantly limits the effectiveness of these systems in reducing road accident fatalities.

To overcome these limitations, there is a need for a cost-effective, energy-efficient, and scalable smart helmet system that integrates real-time accident detection, alcohol monitoring, helmet usage verification, and instant emergency alert transmission. The use of IoT-enabled embedded systems combined with GPS and GSM communication offers a practical solution for reliable operation without dependence on continuous high-speed internet connectivity.

This project focuses on the design and implementation of an IoT-based Smart Helmet system that addresses the shortcomings of existing safety solutions by providing accurate accident detection, prevention of drunk riding, real-time location tracking, and automated emergency notification. The proposed system aims to enhance rider safety, reduce emergency response time, and promote responsible riding behavior, making it suitable for deployment across both urban and rural regions of Nepal.

1.3 Objectives

Primary Objective:

- To develop a smart helmet system capable of detecting accidents and sending real-time emergency alerts.

Secondary Objectives:

1. To integrate an alcohol detection system that prevents the vehicle from starting if the rider is under the influence.
2. To implement a helmet ignition interlock system that ensures the rider wears the helmet before starting the vehicle.
3. To design a cost-effective and user-friendly smart helmet prototype.

2. LITERATURE REVIEW

Motorcycle accidents continue to be a major public safety concern worldwide, particularly in developing countries, where two-wheelers are among the most commonly used modes of transportation. According to the World Health Organization (2023), over 60% of road traffic fatalities in low- and middle-income countries involve two-wheeler riders, emphasizing the urgent need for effective safety solutions. Helmets, as the primary protective gear for riders, play a critical role in preventing serious injuries and fatalities during road accidents. Traditionally, helmets have provided passive protection by absorbing impact energy. However, with the advent of the Internet of Things (IoT) and embedded systems, modern helmets are increasingly being designed to offer active safety features, including accident detection, preventive alerts, rider health monitoring, and realtime emergency communication.

Several research studies have attempted to explore and implement these smart helmet technologies, with varying degrees of success.

[1]**Chandran et al. (2016)** introduced the Konnect smart helmet using accelerometers, Wi-Fi, and cloud services for accident detection and notification. Although it provided realtime alerts, reliance on internet connectivity could affect performance in low-network areas and lacked preventive features. Our project improves reliability using direct GSM messaging and includes helmet usage and alcohol detection for complete rider safety

[2]**Impana H. C, Hamsaveni, M. and Chethana, H.T (2020)** surveyed IoT-based smart helmet systems, comparing microcontroller-based and Raspberry Pi-based approaches. They concluded that camerabased systems improve detection accuracy but increase cost and complexity. However, the study did not present a practical integrated model. Our project addresses this by implementing a compact, low-cost ESP32-based system with efficient sensor integration suitable for real-world deployment.

[3]**S. A. Shabbeer et al. (2017)** proposed a smart helmet for accident detection using an accelerometer and GSM-based alert system. While effective for basic accident notification, the system depended on single-sensor threshold detection, leading to possible false

triggers. Our project enhances detection accuracy through sensor fusion and verification mechanisms.

[4]**Anjaneyulu et al. (2025)** designed an IoT-enabled helmet with an MQ-3 alcohol sensor and buzzer alert system. The system focused on detecting alcohol presence and warning the rider. However, it did not integrate full accident detection or automatic emergency communication. Our project extends this concept by combining alcohol monitoring with accident detection, GPS tracking, and real-time GSM-based emergency alerts.

[5]**Manjesh and Raj (2014)** developed a smart helmet using GSM and GPS for accident detection based on vibration sensors. The system automatically sent location details after impact detection. However, it relied only on vibration sensing, which could cause false alerts and lacked helmet usage and alcohol monitoring features. Our project improves this by integrating MPU6050-based motion analysis, helmet-wear detection, and alcohol ignition control, ensuring better reliability and preventive safety.

[6]**Sur, A., Jatti, V. K., Awasthi, P., Ghosh, B., and Bhattacharya, T. (2026)** presented a systematic review of smart helmet technologies, highlighting the widespread use of accelerometers, alcohol sensors, and wireless communication modules such as Bluetooth, Zigbee, and Wi-Fi. The study showed the growing adoption of low-cost processors like NodeMCU for IoT-based helmet systems. However, the paper mainly analyzed existing technologies and did not propose an optimized real-time safety system. Our project improves this by implementing an ESP32-based smart helmet with efficient sensor integration and fast ESP-NOW communication, providing a practical, low-cost, and reliable real-time accident detection and safety solution.

[7]**Dhulavvagol, P. M., Shet, R., Nashipudi, P., Meti, A. S., and Ganiger, R. (2017)** proposed a smart helmet using GPS, GSM, cloud storage, and an MQ-3 alcohol sensor for accident detection and notification. The system stored sensor data in the cloud and achieved about 84% accident detection accuracy. However, dependence on cloud processing may introduce delays, and the reported accuracy indicates possible false detections. Our project improves reliability by using real-time local motion analysis with MPU6050 and fast

ESPNow communication between helmet and receiver units, enabling quicker response without relying on continuous cloud connectivity.

From these studies, it is evident that while IoT-based helmets hold significant potential to enhance rider safety, there are consistent gaps in existing research. Common issues include the lack of biometric health monitoring, reliance on bulky or mechanical sensors, limited energy optimization, and insufficient focus on modularity and affordability. These gaps are particularly critical in countries like Nepal, where cost constraints, infrastructure limitations, and regional conditions demand practical, low-cost, and scalable solutions.

3. SYSTEM OVERVIEW

3.1 Project Overview

3.1.1 Block Diagram Description

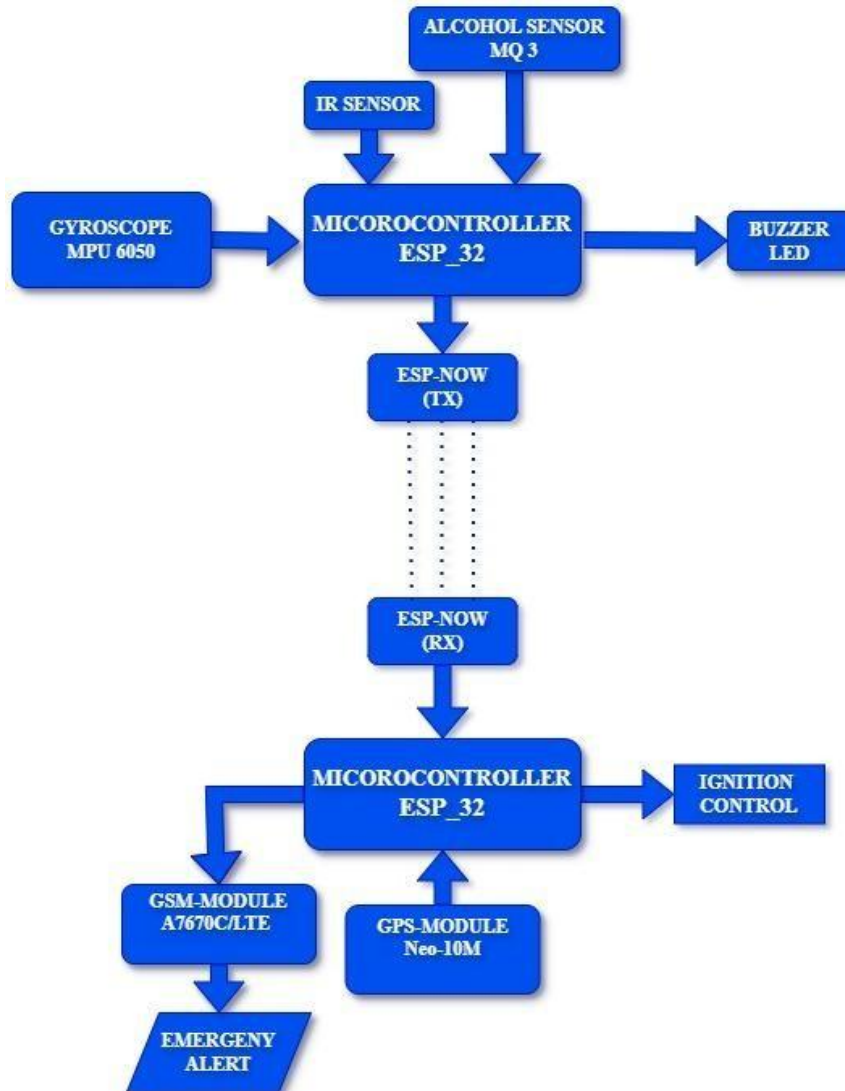


Fig 3.1.1: Block Diagram of smart helmet for accident detection and prevention

3.1.2 Transmitter End

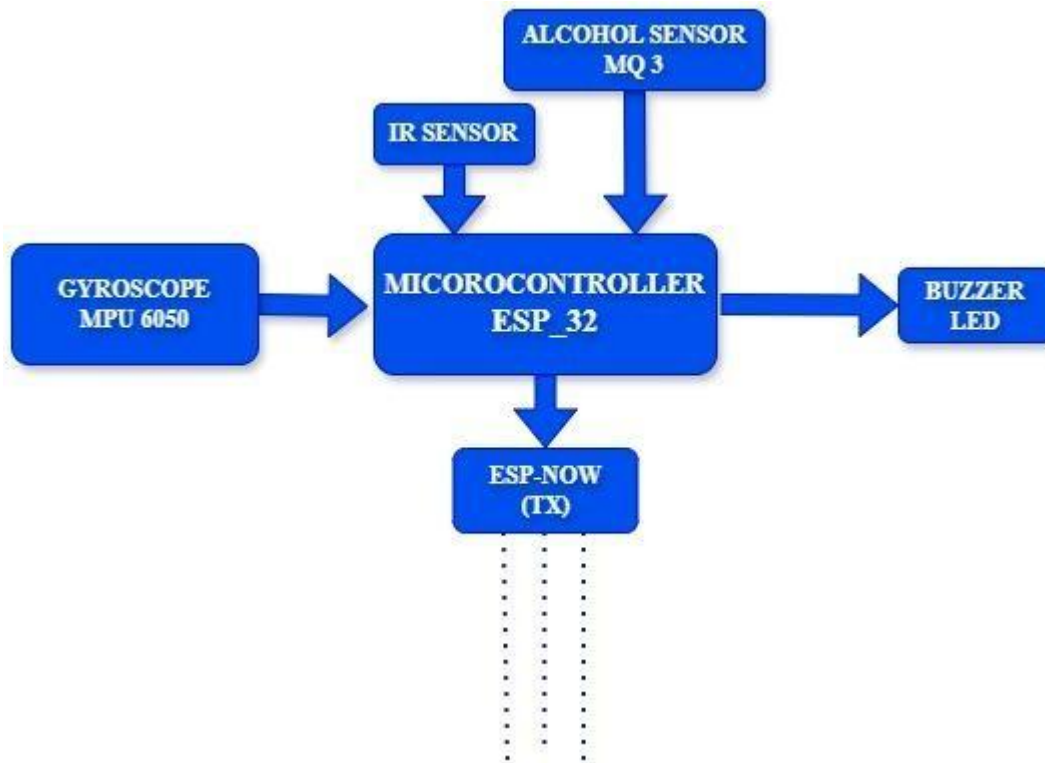


Fig 3.1.2: Transmitter End Block Diagram

The transmitter side of the smart helmet system is designed meticulously to enhance rider safety by detecting accidents, monitoring alcohol consumption, and enforcing helmet usage. The system is a compact integration of multiple critical hardware components, each performing a key role in sensing, processing, and generating real-time safety alerts.

At the center of data acquisition are the integrated sensors. An MQ-3 Alcohol Sensor is employed to detect alcohol in the rider's breath, providing an analog output corresponding to the alcohol concentration. An IR Sensor is used to confirm whether the rider is wearing the helmet, forming the basis of the helmet ignition interlock system. A MPU 6050 Gyroscope Sensor detects sudden impacts or abnormal tilts, enabling accurate accident detection. These sensors continuously provide data to the microcontroller to ensure immediate safety responses.

The ESP32 microcontroller serves as the central processor of the helmet system. It interfaces with all the sensors, acquiring analog data from the MQ-3 and digital signals from the IR and gyroscope sensors. The microcontroller processes this information, applies threshold limits, and triggers appropriate actions such as preventing vehicle ignition if alcohol is detected, or activating a buzzer in case of an accident or helmet removal.

Alerts and notifications are generated via a buzzer, which provides immediate feedback to the rider and surrounding people, ensuring quick response to unsafe conditions. The entire system is powered by a compact battery, guaranteeing reliable operation without dependence on external power sources.

Such a design allows the smart helmet to function as an autonomous, intelligent safety device capable of monitoring critical rider conditions, preventing unsafe vehicle operation, and alerting both the rider and nearby individuals in real time, thereby significantly enhancing road safety.

3.1.3 Receiver End

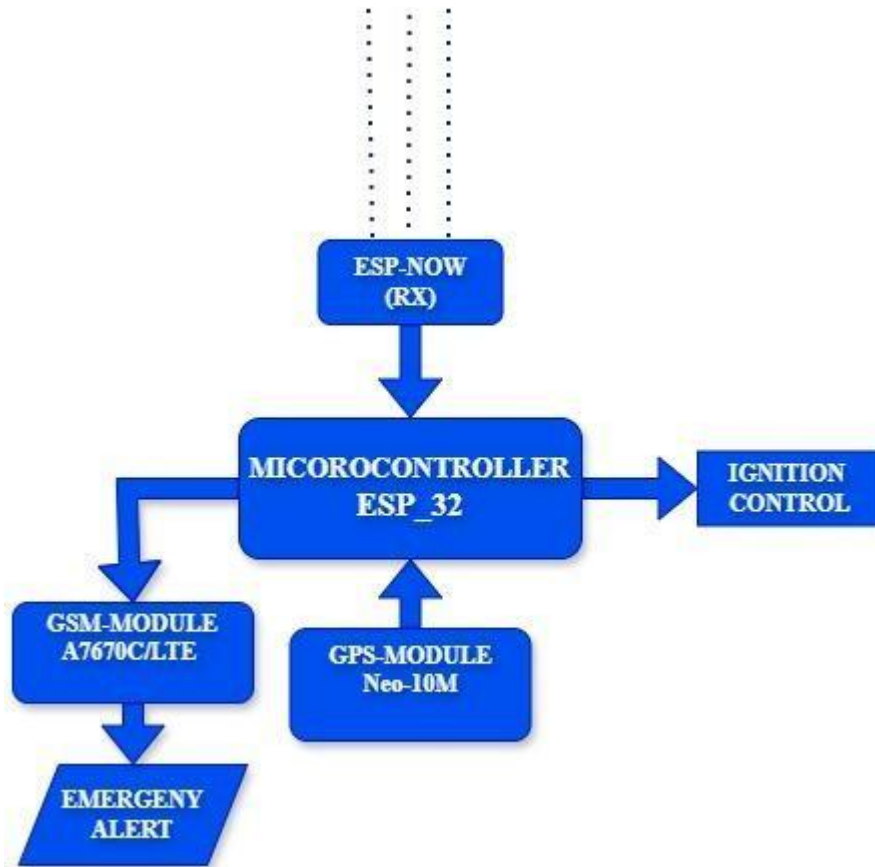


Fig 3.1.3: Receiver End Block Diagram

The receiver side of the smart helmet system is designed to receive real-time safety data from the helmet and immediately alert relevant parties in case of emergencies. The system is an integrated setup of key hardware components, each performing a vital role in data reception, processing, location tracking, and emergency communication.

At the heart of the receiver system is the ESP32 microcontroller, which functions as the central processor. It receives transmitted data from the helmet using the ESP-NOW wireless protocol, allowing low-latency and reliable communication between the helmet (transmitter) and the receiver. The microcontroller interprets incoming signals related to helmet usage, alcohol detection, and accident events.

A GPS Module (NEO-10M) is incorporated to determine the real-time location of the rider when an emergency event is detected. This geolocation information is essential for directing help promptly to the exact site of the accident.

The GSM Module (SIMA7670C) enables the system to send emergency alerts, such as SMS messages or calls, to pre-configured contacts including family members, emergency services, or nearby responders. These alerts contain critical information such as the nature of the event (accident detected) and the precise GPS location of the rider, ensuring a rapid and accurate response.

The receiver system is powered by a dedicated battery, providing uninterrupted operation even in outdoor or mobile scenarios. Together, this setup allows the smart helmet system to function as a complete safety network, capable of monitoring rider status, processing data in real time, and transmitting emergency alerts immediately to reduce response time and enhance road safety.

4. RELEVANT THEORIES

4.1 ESP 32

The ESP32 is a highly versatile, powerful, and cost-effective microcontroller that has become one of the most popular choices for IoT (Internet of Things) and embedded system applications. Developed by Espressif Systems, the ESP32 integrates both Wi-Fi and Bluetooth connectivity within a single chip, eliminating the need for additional communication modules and reducing overall system cost and complexity. Unlike traditional microcontrollers that require external hardware for wireless communication, the ESP32 provides built-in networking capabilities, making it ideal for smart and connected devices. It features a dual-core processor that can operate at speeds of up to 240 MHz, enabling fast data processing and multitasking. With multiple GPIO pins, analog-to-digital converters (ADC), digital-to-analog converters (DAC), PWM channels, SPI, I2C, UART, and other communication interfaces, the ESP32 supports seamless integration with a wide range of sensors and peripherals. This combination of processing power and connectivity makes it highly suitable for real-time embedded applications.

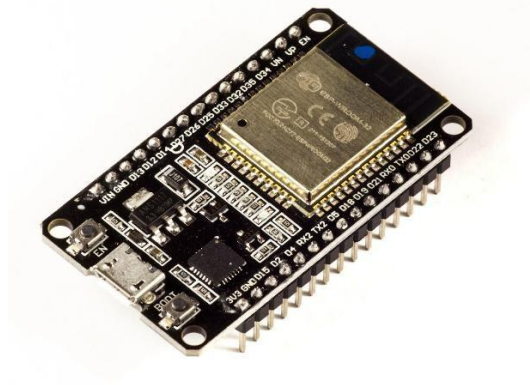


Fig 4.1.1: ESP 32

In the smart helmet system, the ESP32 acts as the central processing unit, managing and coordinating all connected sensors and modules efficiently. It continuously collects and

processes data from components such as the accelerometer, gyroscope, alcohol sensor, and helmet-wearing detection sensor. The high processing capability allows it to analyze sensor data in real time to detect critical events such as accidents, sudden impacts, rider intoxication, or improper helmet usage. Based on the processed data, the ESP32 can trigger alerts, transmit information wirelessly, or activate safety mechanisms as required. Additionally, the ESP32 is designed for low power consumption, which is crucial for wearable and battery-operated devices like a smart helmet. Its compact size, energy efficiency, and reliable performance ensure extended operational time while maintaining portability and user comfort. These features make the ESP32 an ideal microcontroller for implementing advanced safety, monitoring, and communication functions in modern smart helmet systems.

4.2 MQ-3

The MQ-3 alcohol sensor is a gas-detecting component used in the smart helmet system to determine whether the rider has consumed alcohol before operating the vehicle. It is specifically designed to detect ethanol vapors in human breath, making it suitable for breath-analyzer and drunk-driving prevention applications. The sensor works on the principle of change in electrical resistance. It contains a sensing layer made of tin dioxide (SnO_2), which is highly sensitive to alcohol molecules. When alcohol vapors come into contact with the sensing surface, the resistance of the material decreases, producing an analog voltage output proportional to the alcohol concentration. This analog signal is read by the ESP32 microcontroller through its Analog-to-Digital Converter (ADC) pin, where it is converted into a digital value for further processing and comparison with a predefined safe threshold level programmed in the system.

In the smart helmet project, the MQ-3 sensor is positioned near the rider's mouth and nose area to ensure accurate detection of breath alcohol content. Before allowing vehicle ignition, the ESP32 checks the alcohol level detected by the sensor. If the measured value is within the safe limit, the system permits the activation of the ignition relay; however, if the value exceeds the preset threshold, the microcontroller identifies the rider as intoxicated

and prevents the engine from starting by keeping the relay OFF. A warning signal is also transmitted to the bike unit using ESP-NOW wireless communication. The sensor requires a short warm-up time after power-up to stabilize its internal heating element for accurate readings. Once stabilized, it provides quick and reliable detection, helping to reduce drunkdriving incidents and enhance overall road safety.



Fig 4.2.1: MQ3 Sensor

4.3 MPU6050

The MPU6050 is a six-axis motion tracking sensor module that integrates a 3-axis accelerometer and a 3-axis gyroscope into a single compact integrated circuit. It is commonly used in robotics, mobile devices, drones, and wearable safety systems to detect motion, orientation, tilt, and vibration. The accelerometer section measures linear acceleration forces acting on the object along the X, Y, and Z directions, which helps determine whether the helmet is stationary, moving, or experiencing a sudden shock. The gyroscope section measures angular velocity or rotational movement around these same three axes, allowing the system to detect rotation, tilt angle, and direction of movement. By combining both sensors, the MPU6050 can provide complete information about position and motion behavior. The module communicates with the ESP32 microcontroller through the I2C communication protocol using SDA and SCL pins, enabling continuous digital data transfer. Due to its small size, low power requirement, and good accuracy, it is very suitable for wearable devices such as a smart helmet where real-time monitoring is necessary.

In our project, the working of the MPU6050 is based on continuously observing the rider's motion and identifying abnormal situations that indicate an accident. While the motorcycle is being driven normally, the sensor produces stable acceleration and angular velocity values, which the ESP32 reads repeatedly at short intervals. The controller compares these values with predefined safe ranges stored in the program. When a crash, fall, or sudden impact occurs, the helmet experiences a rapid change in acceleration and orientation. The accelerometer detects a large force (sudden jerk), and at the same time the gyroscope detects a sharp tilt or rotation. If these values exceed the threshold level for a certain time duration, the ESP32 interprets it as an accident condition rather than normal movement like braking or turning. Immediately after detection, the controller triggers the emergency procedure: it requests the rider's coordinates from the GPS module and then uses the GSM module to send an alert message containing the location to predefined emergency contacts. Therefore, the MPU6050 acts as the main accident detection unit of the smart helmet, ensuring quick identification of crashes and enabling faster rescue response, which can significantly reduce injury severity and improve rider safety.

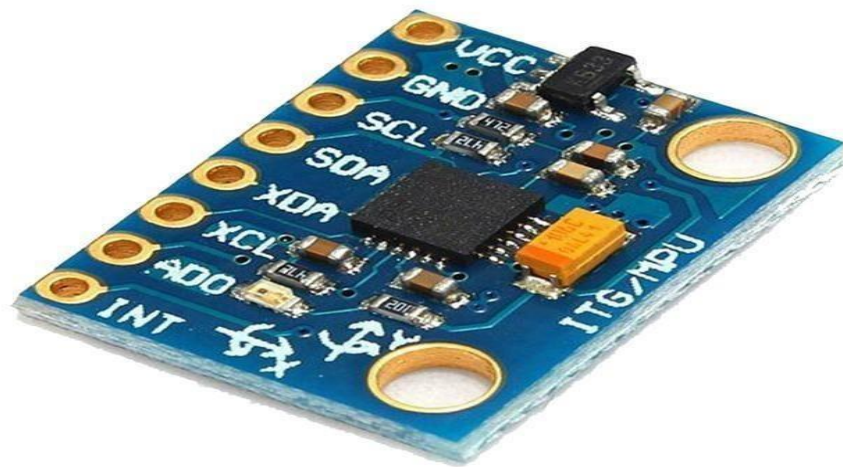


Fig 4.3.1: MPU6050

4.4 A7670C GSM Module

The A7670C GSM module is a cellular communication device used in our smart helmet system to provide wireless connectivity through the mobile network. It works with a standard SIM card and communicates with nearby cellular towers, similar to how a mobile phone operates. The module supports GSM/GPRS communication services such as sending SMS messages, making voice calls, and transmitting small packets of data. It interfaces with the ESP32 microcontroller through serial UART communication using TX (transmit) and RX (receive) pins. The microcontroller controls the module by sending AT commands, which are special instructions used to perform operations like checking network availability, sending text messages, or initiating calls. The module is compact in size, consumes relatively low power, and provides reliable network coverage, making it suitable for embedded IoT safety applications like a wearable smart helmet where dependable long-distance communication is required.



Fig 4.4.1: A7670C GSM Module

In our project, the working of the A7670C module is directly connected to the accident detection and emergency alert system. When the MPU6050 sensor senses a sudden impact or fall, the ESP32 identifies it as an accident condition and immediately requests location data from the GPS module. After receiving the latitude and longitude coordinates, the ESP32 creates an emergency alert message containing an accident warning and a Google Maps location link. The controller then sends a sequence of AT commands to the A7670C module, instructing it to transmit the SMS to predefined emergency contact numbers such

as family members or guardians. Within a few seconds, the message is delivered over the cellular network, allowing the receiver to open the map link and identify the rider's exact location. This process works automatically even if the rider is unconscious and unable to use a phone. Therefore, the A7670C GSM module plays a crucial role in enabling real-time remote communication, reducing rescue time, and significantly improving the safety and effectiveness of the smart helmet system.

4.5 GPS(NEO-10M)

The NEO-10M GPS module is a highly precise and compact satellite positioning device that is widely used in IoT, robotics, and wearable safety systems to provide real-time location tracking. It operates by receiving signals from multiple Global Positioning System (GPS) satellites orbiting the Earth. Using a process called trilateration, the module calculates its precise geographic coordinates, including latitude, longitude, altitude, and sometimes even speed and direction of movement. The NEO-10M communicates with the ESP32 microcontroller via UART serial communication, transmitting data in the form of NMEA (National Marine Electronics Association) sentences, which contain structured information about the position and timing. Its compact size, light weight, and low power consumption make it ideal for wearable devices such as a smart helmet, where continuous operation and minimal bulk are essential. Additionally, it has a fast satellite acquisition time, meaning it can quickly lock onto multiple satellites and provide accurate location data even in outdoor environments with some obstruction.

In our smart helmet project, the NEO-10M GPS module serves a critical role in accident detection and emergency response. During normal operation, the GPS module continuously receives satellite signals and updates the ESP32 with the rider's current location. When the MPU6050 sensor detects a sudden impact or abnormal motion indicating a potential accident, the microcontroller immediately requests the most recent GPS coordinates from the NEO-10M module.



Fig 4.5.1: GPS(NEO-10M)

These coordinates are then processed and formatted into a readable alert message, often including a Google Maps link for convenience. This message is transmitted via the GSM module to predefined emergency contacts, such as family members or guardians, ensuring that they can quickly locate the rider. Because accidents may leave the rider unconscious or unable to call for help, this automatic real-time location tracking is vital for reducing emergency response time. By integrating the NEO-10M GPS module, the smart helmet not only detects accidents accurately but also provides precise location information, which is essential for timely rescue, effective accident management, and overall improvement of rider safety.

4.6 IR SENSOR

The IR (Infrared) sensor is an electronic sensing device widely used for detecting the presence, proximity, or movement of objects without physical contact. It operates by emitting infrared light through an IR LED and measuring the reflected light using a photodiode or phototransistor. When an object comes into the detection range, the emitted infrared light bounces back from the object and is received by the photodiode, producing a measurable electrical signal. This signal can then be interpreted by a microcontroller or processing unit to determine whether an object is present or absent. IR sensors are popular in embedded and IoT systems because they are compact, cost-effective, and capable of

providing fast and reliable detection without requiring any physical touch. They are also energy-efficient, have long operational life, and are not affected by mechanical wear, making them highly suitable for wearable safety devices and automated monitoring applications.



Fig 4.6.1: IR Sensor

In our smart helmet project, the IR sensor is specifically implemented as a helmet-wear detection system, which plays a critical role in ensuring rider safety. The sensor is strategically positioned inside the helmet so that it can detect whether the rider's head is correctly inside the helmet when it is worn. When the rider puts on the helmet, infrared light emitted by the sensor reflects off the rider's head and is received by the photodiode. This reflection generates a signal that is read by the ESP32 microcontroller, indicating that the helmet is properly worn. If the helmet is not worn or is worn improperly, the IR sensor fails to detect any reflection, and the microcontroller keeps the motorcycle ignition disabled through the relay circuit. This automatic detection prevents the vehicle from being started without proper helmet use, thereby enforcing safety compliance. By integrating the IR sensor, the smart helmet system ensures that the safety mechanisms, such as accident detection and emergency alert, are only active when the rider is wearing the helmet, significantly reducing the risk of head injuries and promoting responsible riding habits. Moreover, the IR sensor operates in real time, providing instant feedback to the microcontroller, which helps maintain seamless functioning of the overall safety system, making it a crucial component of the smart helmet design.

4.7 ESP-NOW

ESP-NOW is a proprietary wireless communication protocol developed by Espressif for its ESP32 and ESP8266 microcontrollers. It enables peer-to-peer, low-latency, and lowpower data communication between ESP devices without requiring a Wi-Fi router, hotspot, or internet connection. Operating on the 2.4 GHz Wi-Fi frequency, ESP-NOW allows devices to exchange small data packets quickly and efficiently, making it ideal for realtime applications in IoT, wearable devices, and embedded systems. The protocol supports multiple peers, meaning a single device can communicate with several other devices simultaneously. It is extremely lightweight, consumes minimal power compared to traditional Wi-Fi communication, and allows for distances of approximately 100–200 meters in open environments. ESP-NOW is highly reliable for sending short control or sensor data messages, especially when rapid communication is critical and internet connectivity may be unavailable or unstable.

In our smart helmet project, ESP-NOW is used as the primary wireless communication method between the helmet unit and the motorcycle unit. The helmet houses an ESP32 microcontroller that continuously monitors data from multiple sensors, including the MPU6050 for motion detection, the MQ-3 alcohol sensor for sobriety detection, and the helmet-wear detection sensor. Rather than sending this data over the internet, the helmet ESP32 transmits small packets of safety status information directly to the bike’s ESP32 using ESP-NOW. This direct device-to-device communication ensures near-instantaneous data transfer, which is critical for safety-related decisions. For instance, when the rider wears the helmet correctly, the helmet ESP32 sends a “helmet worn” signal to the motorcycle unit. If alcohol is detected above the safe threshold, a “drunk riding” alert is sent immediately.

The bike-side ESP32 receives these ESP-NOW packets in real time and controls the ignition relay accordingly. Only when both safety conditions — proper helmet usage and no alcohol detected — are satisfied does the relay close the ignition circuit, allowing the motorcycle to start. Any unsafe condition prevents ignition, ensuring the rider cannot operate the vehicle recklessly. ESP-NOW is particularly beneficial for this system because it requires no network configuration, is highly energy-efficient, and has very low

communication latency. By integrating ESP-NOW, our smart helmet project achieves instant, reliable, and wireless communication between the helmet and motorcycle, enabling real-time safety enforcement and significantly reducing the risk of accidents due to helmet non-compliance or drunk riding.



Fig 4.7.1: ESP-NOW Model

4.8 ARDUINO IDE

The Arduino IDE is a user-friendly software platform used for programming microcontrollers, such as the ESP32, which forms the core of the Smart Helmet system. It provides a comprehensive environment to write, compile, and upload code (sketches) to the microcontroller, enabling it to read data from sensors, process the information, and control communication modules like ESP-NOW. The IDE supports numerous pre-built libraries, allowing easy integration of hardware components such as accelerometers, gyroscopes, and GPS modules, which are essential for accident detection and location tracking. It also includes a Serial Monitor, which provides real-time debugging and monitoring of sensor readings and communication status, helping developers test and validate the system effectively. Using the Arduino IDE, the Smart Helmet can be programmed to continuously monitor sensor data, detect abnormal events, and transmit critical information instantly to a receiver device, ensuring reliable, real-time, and low power operation in wearable safety applications.

5. DESIGN & OPERATION

5.1 Flow Chart

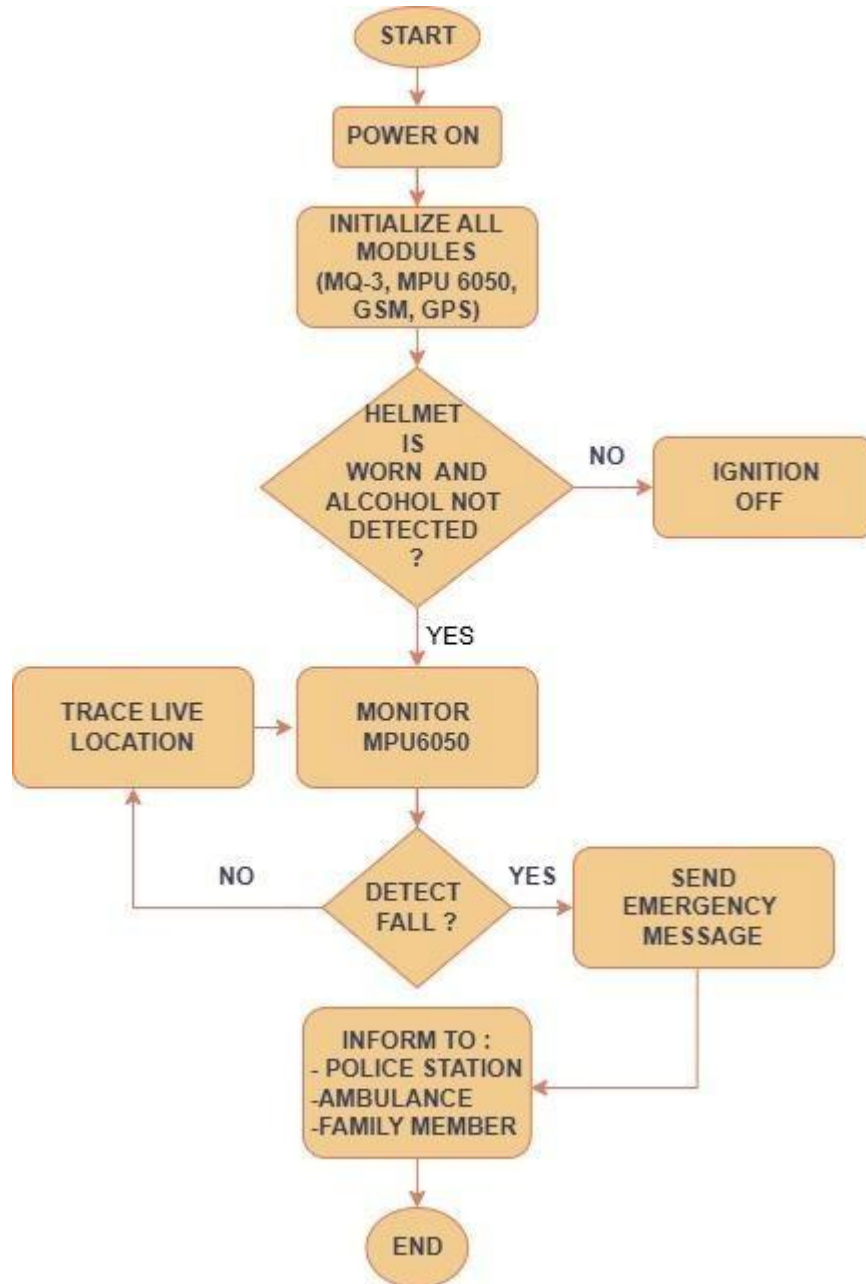


Figure 5.1.1: Flow Chart of smart helmet for accident prevention and detection

5.2 Circuit Diagram

5.2.1 Transmitter End

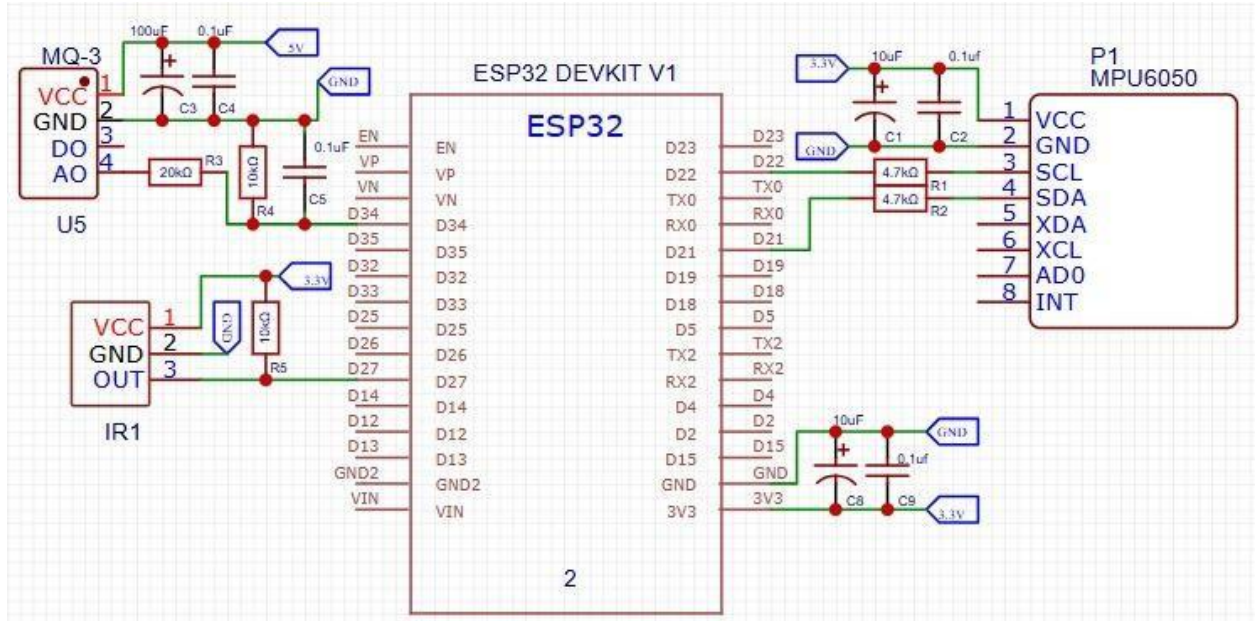


Fig 5.2.1: Circuit diagram for Transmitter side

5.2.2 Receiver End

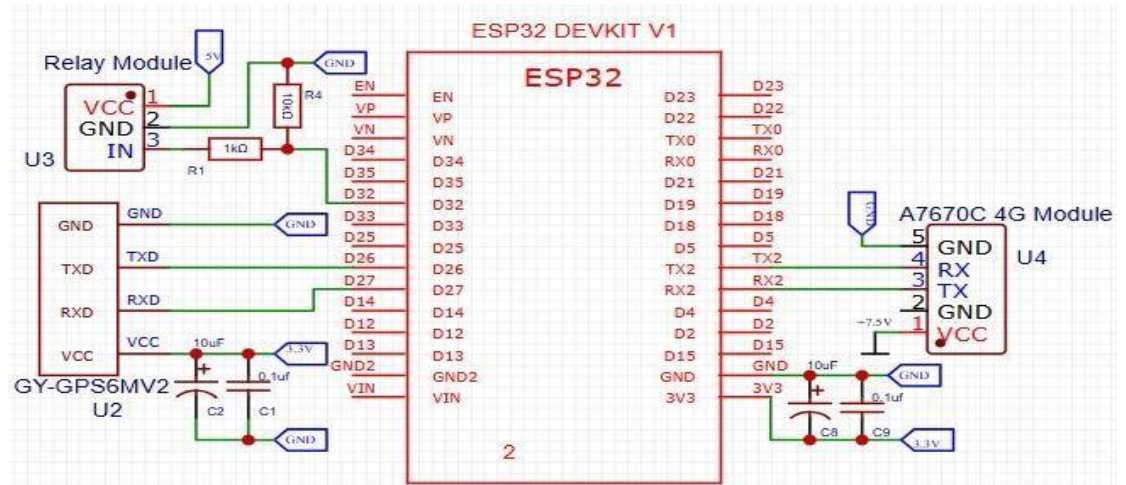


Fig 5.2.2: Circuit diagram for Receiver side

5.3 Hardware Operation

The hardware operation of this Smart Helmet project relies entirely on the properly designed and implemented transmitter and receiver units, which together ensure real-time accident detection, alcohol monitoring, and emergency alerting. The transmitter and receiver PCBs were first designed schematically and with board layout in EasyEDA, followed by careful assembly and soldering of components onto the fabricated boards. This approach provides a compact, durable, and reliable physical realization of the system's electronics, ensuring long-term stability and robustness for wearable use in real-world riding conditions.

On the transmitter side, the helmet PCB is built for autonomous data collection and event detection. It integrates essential sensors, including an IR sensor for obstacle detection, an MQ-3 alcohol sensor for rider sobriety monitoring, and an MPU6050 accelerometer and gyroscope for motion and impact detection. A buzzer provides immediate audio feedback, but it is activated only when the MPU6050 detects a sudden change in motion or orientation, indicating a potential accident. To verify whether a real accident has occurred, the ESP32 continuously monitors for further motion over a period of 20 to 30 seconds. If no significant movement is detected during this interval, the system interprets it as a confirmed accident, and the buzzer remains active only during this verification period. This logic ensures that minor bumps or false triggers do not activate unnecessary alarms while enabling accurate and timely emergency response. All sensors and the buzzer are connected to an ESP32 microcontroller, which serves as the central processing unit. The ESP32 continuously reads sensor data, evaluates thresholds for alcohol, motion, and collisions, and formats the data for transmission. ESP-NOW serves as the main communication protocol, providing low-latency, secure, peer-to-peer wireless transmission of small encrypted data packets to the receiver side. The transmitter PCB is powered by a 3.7V Liion battery, with onboard voltage regulation ensuring stable power to both the ESP32 and all connected sensors. Component placement and trace routing were optimized to minimize electrical noise and maintain reliable signal integrity, making the helmet's electronics compact and robust.

Meanwhile, the receiver side contains a dedicated PCB designed to receive, process, and forward alerts from the helmet. It uses an ESP32 microcontroller to receive ESP-NOW

data packets from the helmet and interprets sensor readings for critical events. A NEO10M GPS module determines the helmet's real-time location whenever an alert is received, while the A7670C 4G GSM module enables sending SMS notifications or emergency alerts to predefined contacts. Additionally, a relay module can control external devices, such as vehicle safety mechanisms or warning alarms. The receiver PCB is powered by a regulated battery supply, ensuring stable operation of all modules. The design emphasizes reliable electrical connections, minimal noise interference, and smooth operation of high-speed data communication and peripheral control.

The interaction between the transmitter and receiver is driven entirely by ESP-NOW, which ensures instantaneous, low-power, and secure peer-to-peer communication. When the helmet detects an accident, alcohol presence, or proximity hazard, the transmitter ESP32 sends the data packet immediately to the receiver ESP32. The receiver then activates GPS tracking, sends emergency messages via GSM, and triggers any connected relays or alarms, providing real-time monitoring and instant emergency response.

5.4 Software Operation

The software aspect of the Smart Helmet project serves as the central intelligence, responsible for converting raw sensor data from the helmet into actionable information, managing communication, and enabling real-time emergency responses. This software framework operates across both the helmet-mounted transmitter and the receiver unit, creating a continuous data pipeline from accident or hazard detection to notification and control actions.

On the transmitter side, the ESP32 microcontroller hosts embedded firmware that has been meticulously developed to manage all local operations. The firmware continuously monitors sensors including the MPU6050 accelerometer and gyroscope for sudden motion or impacts, the MQ-3 alcohol sensor for rider sobriety, and the IR sensor for obstacle detection. The firmware implements logic to evaluate whether the readings exceed predefined safety thresholds. When a sudden impact is detected by the MPU6050, the firmware triggers the buzzer to alert the rider immediately. A motion verification routine is also employed: the ESP32 monitors for continued motion over 20–30 seconds to

distinguish minor bumps from real accidents. All sensor readings, including helmet status, alcohol detection, and accident status, are packaged into a lightweight structured data format. This data is then transmitted via ESP-NOW, a low-latency, secure, peer-to-peer communication protocol, to the receiver unit. The firmware also manages power to the system, ensuring efficient use of the 3.7V Li-ion battery, which allows the helmet to remain functional for extended periods.

On the receiver side, another ESP32 microcontroller handles incoming data packets from the helmet via ESP-NOW. The software parses the structured messages and evaluates the information for hazard conditions. If the helmet is being worn and no alcohol is detected, the relay module is controlled to allow vehicle ignition; otherwise, ignition is disabled for safety. In the case of a confirmed accident, the software triggers the NEO-10M GPS module to obtain real-time coordinates of the rider and uses the A7670C 4G GSM module to send an emergency SMS containing accident details and a Google Maps link to predefined contacts. The receiver software also ensures reliable relay operation and realtime logging of received events, providing both automated safety responses and a record of incidents.

The software architecture is modular and event-driven, separating tasks such as sensor reading, motion verification, buzzer control, communication, GPS reading, and GSM messaging. This modularity ensures that each component operates efficiently and reliably without blocking other critical processes. For example, while waiting for GPS coordinates or SMS confirmation, the system continues to monitor incoming ESP-NOW packets, ensuring real-time responsiveness in all situations.

6. CONCLUSION

6.1 Application

1. Accident Detection & Emergency Alerts: Detects falls or collisions and sends real-time SMS with GPS location to emergency contacts.
2. Alcohol Monitoring & Vehicle Safety: Prevents vehicle ignition if the rider is under the influence, reducing drunk driving risks.
3. Preventive Safety Monitoring: Tracks sudden tilts or impacts to distinguish minor bumps from real accidents.
4. Helmet Usage Verification: Ensures the vehicle cannot start if the helmet is not worn, promoting rider safety compliance.
5. Buzzer Alerts for Immediate Response: Provides audible message for the sudden fall or change in the motion to ensure the accident occurs or not.

6.2 Challenges and Difficulties Faced

During the development of the Smart Helmet project, one of the primary challenges was sensor calibration and accuracy. The MPU6050 accelerometer and gyroscope, responsible for detecting sudden motion and impacts, required precise calibration to differentiate between minor bumps and real accidents. Similarly, the MQ-3 alcohol sensor had to be carefully adjusted to ensure reliable detection without triggering false positives due to environmental conditions. Achieving accurate readings from all sensors simultaneously was critical for the safety and reliability of the system.

Another major difficulty was power management. Since the helmet needed to operate continuously on a compact 3.7V Li-ion battery, the software had to manage energy efficiently, especially while handling real-time monitoring and wireless communication via ESP-NOW. Maintaining a balance between system responsiveness and battery life required careful optimization of both firmware routines and sensor polling intervals.

Hardware integration posed additional challenges. Combining multiple components—including the sensors, buzzer, relay, GPS, and GSM modules—onto a compact PCB required careful layout planning, trace routing, and mitigation of electrical noise and interference.

Ensuring reliable operation of each module while maintaining the helmet's ergonomic and lightweight design was a significant concern.

Software logic and system synchronization were also difficult to implement effectively. The motion-verification routine had to confirm accidents by monitoring for no significant motion over 20–30 seconds, preventing false alarms. Synchronizing GPS and GSM modules to send timely alerts with accurate location data during an accident event required precise programming and testing. Despite these difficulties, iterative testing, careful optimization, and modular system design resulted in a reliable, wearable, and functional Smart Helmet capable of real-time accident detection and rider safety management.

6.3 Successful Outcome

Accident Detection: The MPU6050 accelerometer and gyroscope detect sudden impacts, abnormal tilts, and falls by continuously monitoring motion changes.

Alcohol Monitoring: The MQ-3 sensor detects alcohol vapors from the rider's breath and prevents vehicle ignition if the alcohol level exceeds the predefined limit.

Helmet Wearing Detection (IR Sensor): The IR sensor verifies whether the helmet is worn by detecting infrared reflection from the rider's head. If the helmet is not worn, the vehicle ignition system remains disabled.

Low-Latency Communication: ESP-NOW enables secure and real-time peer-to-peer communication between the helmet and receiver unit.

Emergency Alerts: In the event of a confirmed accident, the receiver sends GPS-enabled SMS notifications to predefined emergency contacts.

Buzzer and Motion Verification: Audible buzzer alerts combined with motion verification logic help differentiate real accidents from minor disturbances.

Efficient Power Management: The system operates reliably on a 3.7V Li-ion battery with optimized power consumption.

Compact and Wearable Design: Careful PCB integration and component placement ensure a lightweight, ergonomic, and user-friendly helmet design.

Automated Safety Control: The receiver automatically manages vehicle ignition and emergency response actions to enhance overall rider safety.

6.6 Future Enhancement

- Cloud Connectivity: Enable real-time data storage and remote monitoring via a mobile app or web platform.
- Fleet Monitoring: Track multiple riders simultaneously and analyze riding patterns for enhanced safety.
- Advanced Sensors: Integrate additional sensors like air quality, temperature, or cameras for environmental monitoring and accident recording.
- Voice Alerts: Provide hands-free voice warnings to alert riders more intuitively.
- Battery Optimization: Use energy-efficient components or integrate solar charging to extend battery life.
- Miniaturization: Further reduce helmet size and weight for better ergonomics and comfort.
- Biometric Monitoring: Add heart rate, body temperature, or SpO₂ sensors to track rider health and detect fatigue or medical emergencies.

6.5 Assumptions

Critical assumptions have to be made in order to successfully implement our project. The list of assumptions for our project are:

- Sensor Reliability: MPU6050, MQ-3, and IR sensors provide reliable and accurate data after proper calibration.
- Battery Performance: The 3.7V Li-ion battery provides sufficient power for continuous operation during normal usage.

- Network Availability: GSM network and GPS signals are available at the time of an accident for sending location-based SMS alerts.
- Communication Stability: ESP-NOW communication between the helmet and receiver remains stable within the operational range.
- Proper Installation: The helmet system and vehicle ignition connections are properly installed and configured.
- User Compliance: The rider wears the helmet correctly and maintains the device in functional condition.
- Physical Durability: The hardware components are reasonably protected from physical damage, environmental conditions, and tampering.

7. Budget Structure

S.N	Components	Qty.	Price per Unit	Total Price
1.	ESP32	2	850	1700
2.	IR Sensor	1	140	140
3.	MQ3 Sensor	1	400	400
4.	MPU 6050	1	520	520
5.	GPS NEO-10M module	1	2750	2750
6.	GSM A7670C module	1	3450	3450
7.	Relay	1	60	60
8.	Buzzer	1	50	50
9.	Battery Holder	2	150	300
10.	PCB Board	2	50	100
11.	Miscellaneous		2000	2000
TOTAL				11,470

8. Working Schedule

Task	June	July	Aug	Sept	Oct	Nov	Dec	Jan	Feb
Problem Analysis									
Proposal Writing									
Presentation and Review									
Component Collection									
Circuit Design & Coding									
Circuit Implementation									
Software Implementation									
Testing and Debugging									
Result Evaluation									
Report Writing									

9. References

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- [7] P. Dhulavvagol, R. Shet, P. Nashipudi, A. S. Meti and R. Gangir, "Smart helmet with cloud GPS GSM technology for accident and alcohol detection," *international conference on cognetive computing and information processing* , pp. 346-357, 2017.

10. APPENDIX

10.1. Datalogger for transmeter

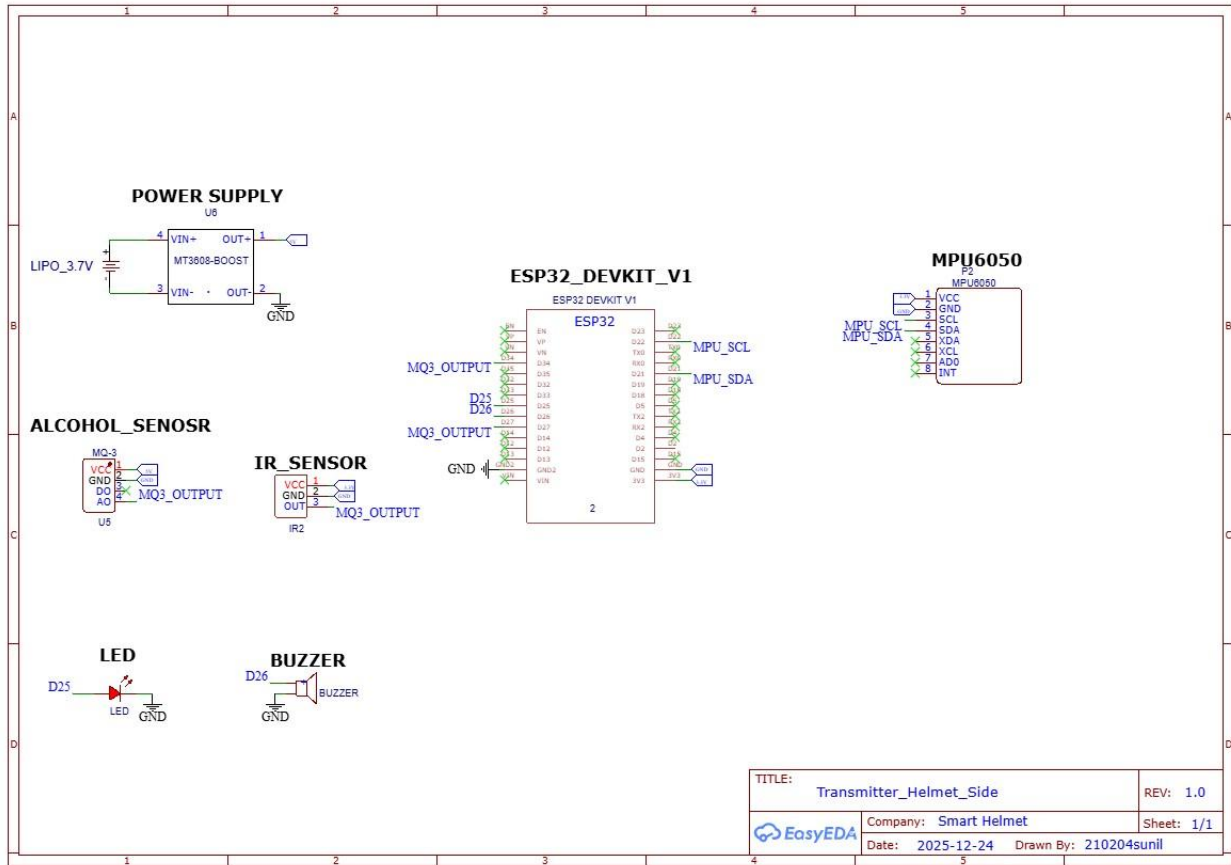


Fig 10.1.1: Datalogger of Transmitter side

10.2 Datalogger for Receiver

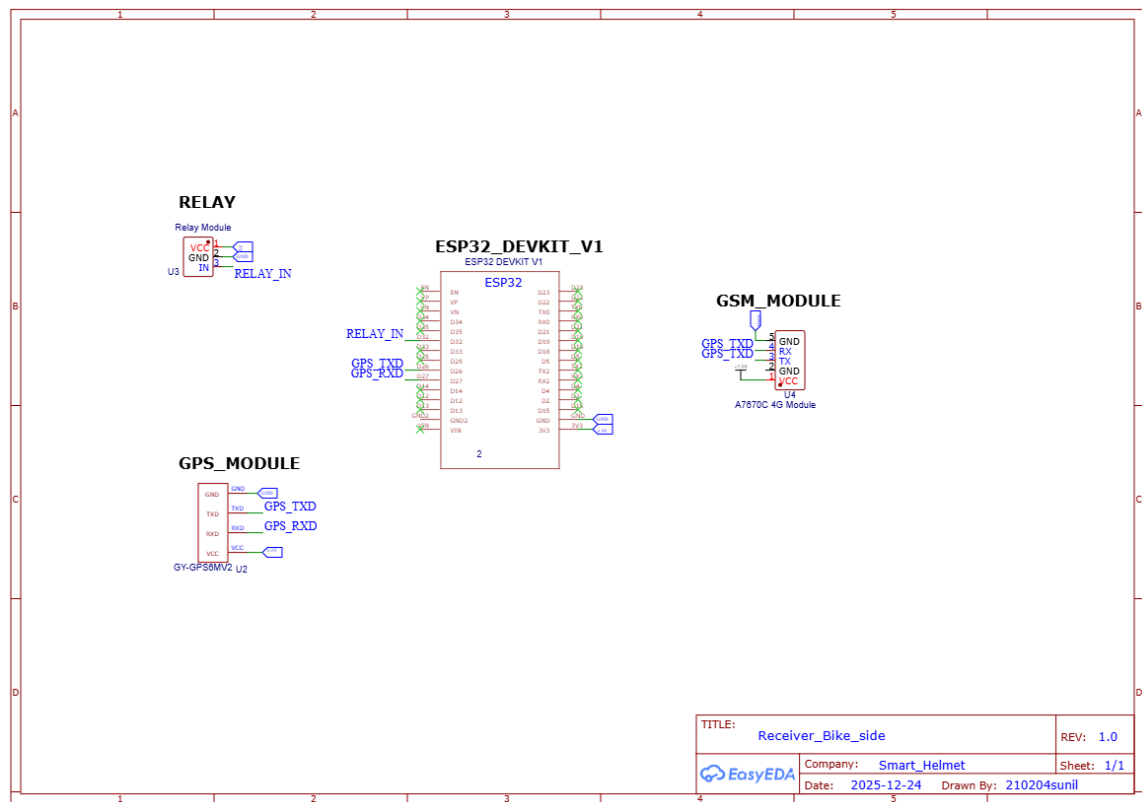


Fig 10.2.1: Datalogger of Receiver side

10.3 Prototype module photos



Fig 10.3.1: Real Time Connection of Transmitter End

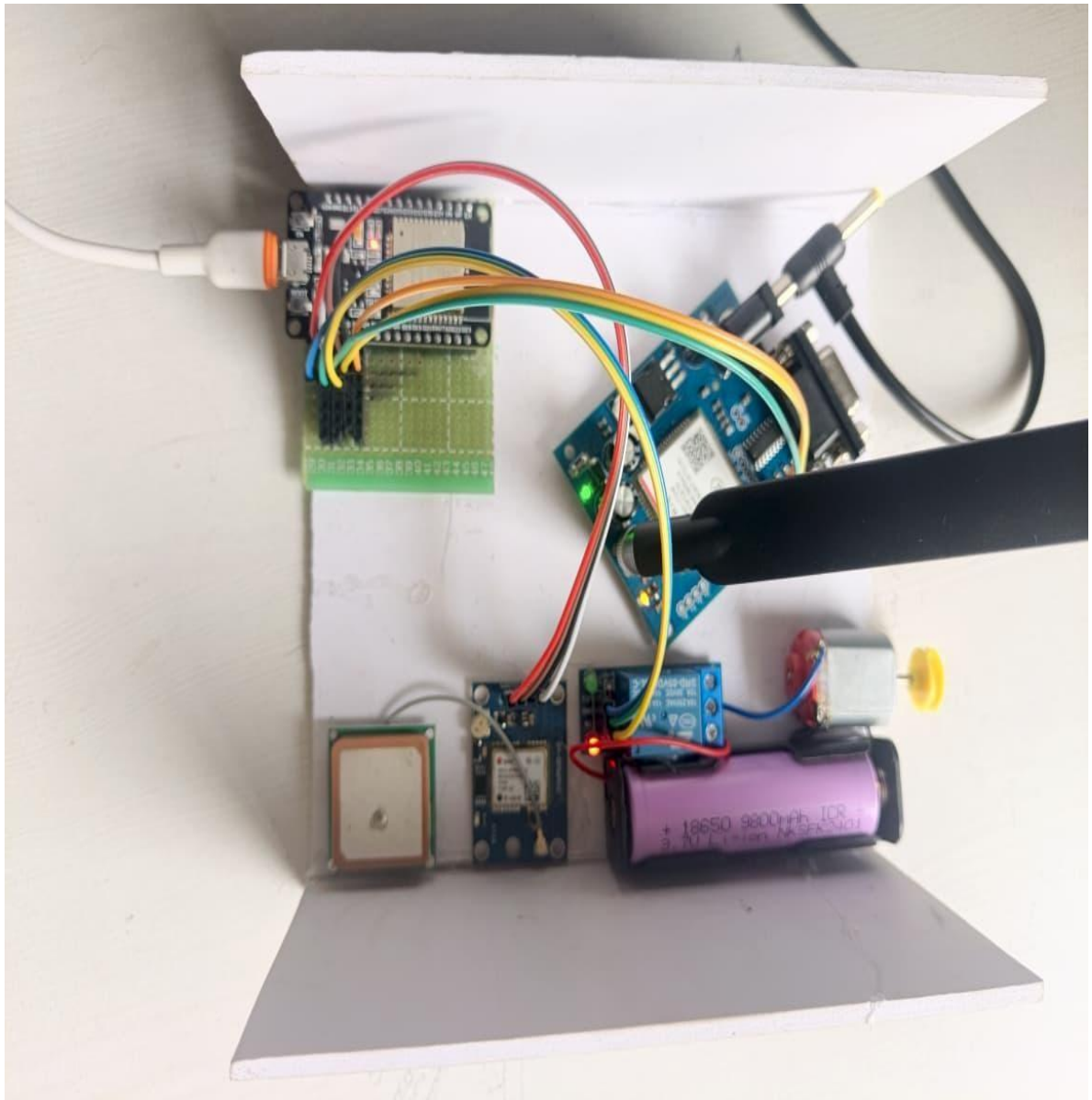


Fig 10.3.2: Real Time Connection of Receiver End

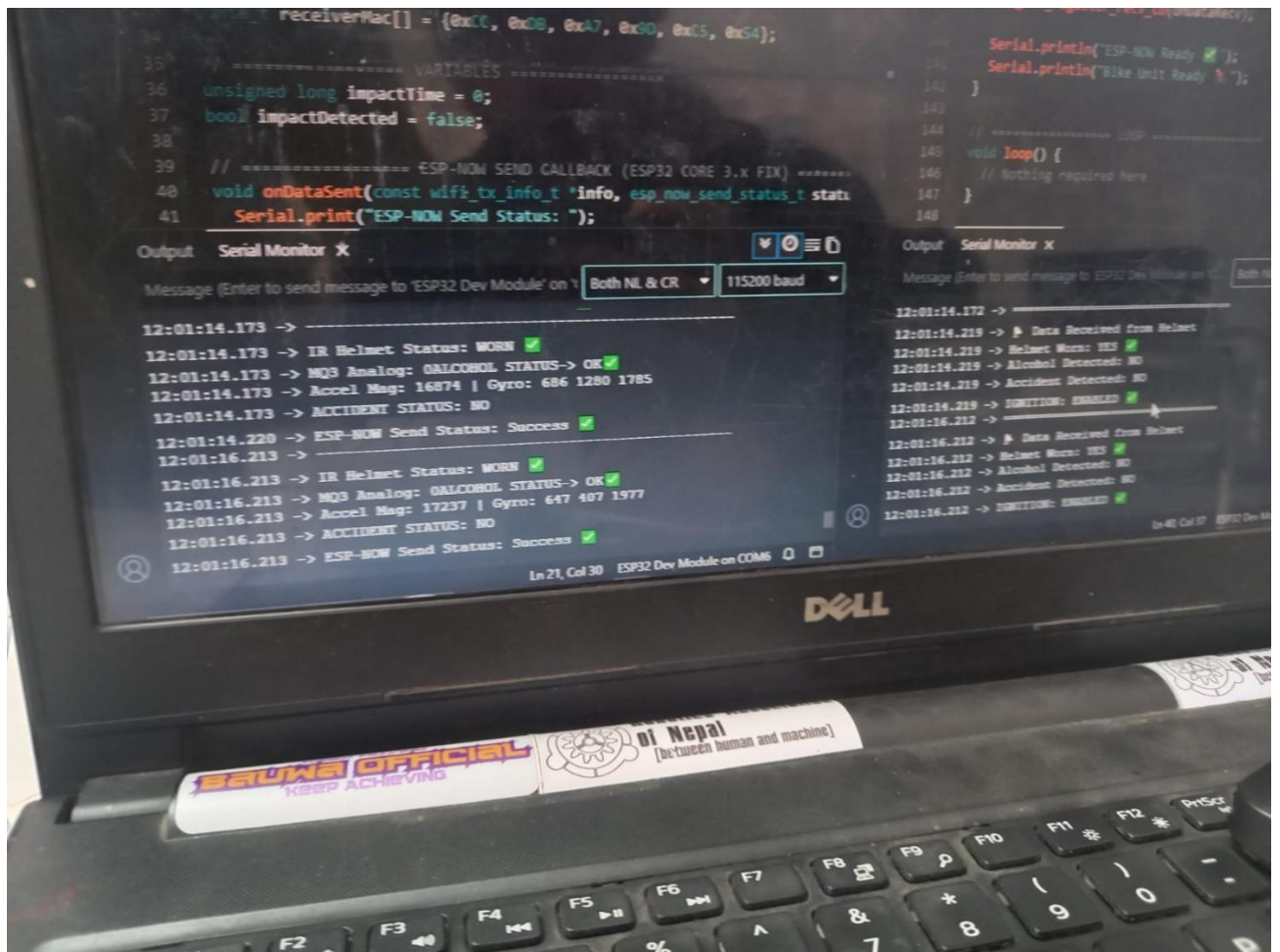


Fig 10.3.3: Tested Outcome

10.4 Code For Transmitter

```
#include <Wire.h>
#include <MPU6050.h>
#include <esp_now.h>
#include <WiFi.h>
#include <math.h>

// ===== MPU6050 =====
MPU6050 mpu(0x68);

// ===== PIN DEFINITIONS =====
#define IR_PIN      27
#define MQ3_ANALOG_PIN  34
#define MQ3_DIGITAL_PIN 14
#define BUZZER_PIN    26
#define ALCOHOL_LED    25

// ===== THRESHOLDS =====
#define ACC_THRESHOLD  30000
#define GYRO_THRESHOLD 20000
#define ALCOHOL_LIMIT  2500
#define NO_MOVEMENT_TIME 5000

// ===== ESP-NOW STRUCT =====
typedef struct struct_message {
    bool helmetWorn;
    bool alcoholDetected;
    bool accidentDetected;
```

```

} struct_message;

struct_message helmetData;

// 🔄 Replace with your receiver ESP32 MAC
uint8_t receiverMac[] = {0xCC, 0xDB, 0xA7, 0x9D, 0xC5, 0x54};

// ===== VARIABLES =====

unsigned long impactTime = 0;

bool impactDetected = false;

// ===== ESP-NOW SEND CALLBACK (FIXED FOR YOUR VERSION) =====

void onDataSent(const uint8_t *mac_addr, esp_now_send_status_t status) {
    Serial.print("ESP-NOW Send Status: ");

    Serial.println(status == ESP_NOW_SEND_SUCCESS ? "Success ✔" : "Fail ✖");

    Serial.print("Sent to: ");

    for (int i = 0; i < 6; i++) {
        Serial.printf("%02X", mac_addr[i]);

        if (i < 5) Serial.print(":");
    }

    Serial.println();
}

void setup() {
    Serial.begin(115200);

    delay(2000);

    // ===== PIN INIT =====

    pinMode(IR_PIN, INPUT);

    pinMode(MQ3_DIGITAL_PIN, INPUT);
}

```

```

pinMode(BUZZER_PIN, OUTPUT);
pinMode(ALCOHOL_LED, OUTPUT);
digitalWrite(BUZZER_PIN, LOW);
digitalWrite(ALCOHOL_LED, LOW);
// ===== I2C INIT =====
Wire.begin(21, 22);
// ===== MPU INIT =====
Serial.println("Initializing MPU6050...");
mpu.initialize();
if (!mpu.testConnection()) {
    Serial.println("MPU6050 NOT connected ✖");
    while (1);
}
Serial.println("MPU6050 connected ✔");
// ===== WIFI INIT =====
WiFi.mode(WIFI_STA);
// ===== ESP-NOW INIT =====
if (esp_now_init() != ESP_OK) {
    Serial.println("ESP-NOW Init Failed ✖");
    return;
}
esp_now_register_send_cb(onDataSent);
esp_now_peer_info_t peerInfo = {};
memcpy(peerInfo.peer_addr, receiverMac, 6);
peerInfo.channel = 0;

```

```

peerInfo.encrypt = false;
if (esp_now_add_peer(&peerInfo) != ESP_OK) {
    Serial.println("Failed to add peer ✖");
    return;
}

Serial.println("ESP-NOW Initialized ✔");
Serial.println("Warming MQ-3 Sensor...");
delay(30000); // MQ-3 warm-up
}

void loop() {
    // ===== IR SENSOR =====
    int irValue = digitalRead(IR_PIN);
    helmetData.helmetWorn = (irValue == LOW);
    // ===== MQ-3 SENSOR =====
    int mq3Analog = analogRead(MQ3_ANALOG_PIN);
    helmetData.alcoholDetected = (mq3Analog > ALCOHOL_LIMIT);
    // ===== ALCOHOL LED =====
    digitalWrite(ALCOHOL_LED, helmetData.alcoholDetected);
    // ===== MPU6050 =====
    int16_t ax, ay, az, gx, gy, gz;
    mpu.getMotion6(&ax, &ay, &az, &gx, &gy, &gz);
    long accMagnitude = sqrt((long)ax * ax +
                             (long)ay * ay +
                             (long)az * az);
    // Impact detection

```

```

if (accMagnitude > ACC_THRESHOLD ||
    abs(gx) > GYRO_THRESHOLD ||
    abs(gy) > GYRO_THRESHOLD ||
    abs(gz) > GYRO_THRESHOLD) {
    impactTime = millis();
    impactDetected = true;
}

if (impactDetected && (millis() - impactTime > NO_MOVEMENT_TIME)) {
    helmetData.accidentDetected = true;
    impactDetected = false;
} else {
    helmetData.accidentDetected = false;
}

// ===== BUZZER =====
digitalWrite(BUZZER_PIN, helmetData.accidentDetected);

// ===== SERIAL OUTPUT =====
Serial.println("-----");
Serial.print("Helmet: ");
Serial.println(helmetData.helmetWorn ? "WORN ✓" : "NOT WORN ✗");
Serial.print("MQ3 Value: ");
Serial.print(mq3Analog);

Serial.println(helmetData.alcoholDetected ? " 🚫 ALCOHOL DETECTED" : "
OK");
Serial.print("Accel Magnitude: ");
Serial.print(accMagnitude);

```

```

Serial.print(" | Gyro: ");
Serial.print(abs(gx)); Serial.print(" ");
Serial.print(abs(gy)); Serial.print(" ");
Serial.println(abs(gz));
Serial.print("ACCIDENT: ");
Serial.println(helmetData.accidentDetected ? "🚗 YES" : "NO");
// ===== SEND DATA =====
esp_now_send(receiverMac, (uint8_t *)&helmetData, sizeof(helmetData));
delay(2000);
}

```

10.5 Code for Receiver

```

#include <WiFi.h>
#include <esp_now.h>
// ===== PIN DEFINITIONS =====
#define RELAY_PIN 32
#define GSM_RX 16
#define GSM_TX 17
#define GPS_RX 27
#define GPS_TX 26
// ===== GSM SERIAL =====
HardwareSerial SIM7670Serial(2); // UART2 for GSM
// ===== STRUCT (Must Match TX Side) =====
typedef struct struct_message {

```

```

bool helmetWorn;
bool alcoholDetected;
bool accidentDetected;
} struct_message;
struct_message receivedData;
// ===== GPS VARIABLES =====
String latitude = "";
String longitude = "";
// ===== RELAY CONTROL =====
void controlIgnition() {
    if (receivedData.helmetWorn && !receivedData.alcoholDetected) {
        digitalWrite(RELAY_PIN, LOW);
        Serial.println("IGNITION: ENABLED ✓");
    } else {
        digitalWrite(RELAY_PIN, HIGH);
        Serial.println("IGNITION: DISABLED ✗");
    }
}
// ===== GPS READ FUNCTION =====
void readGPS() {
    while (Serial1.available()) {
        String line = Serial1.readStringUntil('\n');
        if (line.startsWith("$GPGGA")) {
            int latIndex = line.indexOf(",") + 1;
            latitude = line.substring(latIndex, latIndex + 9);

```

```

        int lonIndex = line.indexOf(",", latIndex + 1) + 1;
        longitude = line.substring(lonIndex, lonIndex + 10);
    }
}

// ===== GSM AT COMMAND FUNCTION =====

void sendATCommand(const char* cmd, const char* expectedResponse, unsigned
long timeout) {
    SIM7670Serial.println(cmd);
    String response = "";
    unsigned long startTime = millis();
    bool responseOK = false;
    while (millis() - startTime < timeout) {
        while (SIM7670Serial.available()) {
            char c = SIM7670Serial.read();
            response += c;
        }
        if (response.indexOf(expectedResponse) != -1) {
            responseOK = true;
            break;
        }
    }
    Serial.println("AT Command Response:");
    Serial.println(response);
    if (responseOK) {

```

```

    Serial.println("Response OK");
} else {
    Serial.println("Timeout without expected response");
}
}

// ===== GSM SEND SMS =====

void sendSMS(String number, String message) {
    SIM7670Serial.print("AT+CMGS=\"");
    SIM7670Serial.print(number);
    SIM7670Serial.println("\"");
    delay(1000); // Wait for >
    SIM7670Serial.print(message);
    delay(500);
    SIM7670Serial.write(26); // CTRL+Z
    delay(5000);
    while (SIM7670Serial.available()) {
        Serial.write(SIM7670Serial.read());
    }
    Serial.println("SMS Sent ✓");
}

// ===== ACCIDENT HANDLER =====

void handleAccident() {
    digitalWrite(RELAY_PIN, HIGH);
    Serial.println("🚗 ACCIDENT DETECTED! Ignition OFF");
    readGPS();
}

```

```

// ----- PRINT GPS DATA TO SERIAL MONITOR -----
Serial.println("----- GPS DATA -----");
Serial.print("Latitude: ");
Serial.println(latitude);
Serial.print("Longitude: ");
Serial.println(longitude);
String mapsLink = "https://maps.google.com/?q=" + latitude + "," + longitude;
Serial.print("Google Maps Link: ");
Serial.println(mapsLink);
Serial.println("-----");
// -----

String alertMsg = "🚗 SMART HELMET ALERT 📶\n";
alertMsg += "Accident Detected!\n";
alertMsg += "Location:\n";
alertMsg += mapsLink;
sendSMS("+9779864757238", alertMsg);
}

// ===== ESP-NOW RECEIVE CALLBACK
=====

void onDataRecv(const esp_now_recv_info_t *info, const uint8_t *incomingData, int len) {
    memcpy(&receivedData, incomingData, sizeof(receivedData));
    Serial.println("=====");
    Serial.println("📶 Data Received from Helmet");
    Serial.print("Helmet Worn: ");

```

```

Serial.println(receivedData.helmetWorn ? "YES ✓" : "NO ✗");
Serial.print("Alcohol Detected: ");
Serial.println(receivedData.alcoholDetected ? "YES 🚗" : "NO");
Serial.print("Accident Detected: ");
Serial.println(receivedData.accidentDetected ? "YES 🚑" : "NO");
controlIgnition();
if (receivedData.accidentDetected) {
    handleAccident();
}
}
// ===== SETUP =====
void setup() {
    Serial.begin(115200);
    delay(2000);
    pinMode(RELAY_PIN, OUTPUT);
    digitalWrite(RELAY_PIN, LOW);
    // GPS
    Serial1.begin(9600, SERIAL_8N1, GPS_RX, GPS_TX);
    // GSM
    SIM7670Serial.begin(115200, SERIAL_8N1, GSM_RX, GSM_TX);
    delay(3000);
    sendATCommand("AT", "OK", 3000);
    sendATCommand("AT+CMGF=1", "OK", 3000);
    sendATCommand("AT+CNMI=1,2,0,0,0", "OK", 3000);
    Serial.println("Initializing WiFi (ESP-NOW Mode)...");
}

```

```

WiFi.mode(WIFI_STA);
if (esp_now_init() != ESP_OK) {
    Serial.println("ESP-NOW Init Failed ✖");
    return;
}
esp_now_register_recv_cb(onDataRecv);
Serial.println("ESP-NOW Ready ✔");
Serial.println("Bike Unit Ready 🚲");
}
// ===== LOOP =====
void loop() {
    // Nothing required here
}

```